Artificial Intelligence

BS (CS) Spring 2026

Supervised Learning

Learning Objectives:

1. Decision Tree

Lab Tasks

Submission Instructions

1. Create one notebook file for all questions with name *ROLLNO_SEC_LAB13* **e.g. i23XXXX_A_LAB013**
2. Submit your .ipynb file with correct name on Google Classroom under your section heading.
3. If you don't follow the above-mentioned submission instruction, you will be marked **zero.**

Plagiarism in the Lab Task will result in **zero** marks in the whole category.

Questions

Question 1: Decision Tree Construction and Interpretation

Dataset to use: Kaggle Purchase Dataset
(https://www.kaggle.com/datasets/akalyasubramanian/dataset-for-decision-tree-classification)

You are required to build a decision tree classifier to predict whether a customer will purchase a product based on features such as age, estimated salary, and gender. The goal of this lab is to understand how decision trees are constructed and how their decisions can be interpreted.

Tasks

1. Load the dataset using Pandas and display the first few rows. Identify the input features and the target variable.
2. Perform data preprocessing:
    - Identify categorical and numerical features.
    - Encode categorical variables using appropriate encoding techniques.
    - Check for missing values and handle them if present.
3. Split the dataset into training and testing sets using a 70:30 ratio.
4. Train two decision tree classifiers using:
    - Gini index
    - Entropy
5. Evaluate both models on the test set using accuracy, precision, recall, and confusion matrix.
6. Visualize one of the trained decision trees and analyze its structure.
7. Select at least two root-to-leaf paths and explain the decision logic in plain language.
8. Compare the performance of both criteria and justify which one performs better for this dataset.

Question 2: Overfitting and Pruning in Decision Trees

Dataset to use: Kaggle Purchase Dataset (https://www.kaggle.com/datasets/akalyasubramanian/dataset-for-decision-tree-classification)

In this lab, you will investigate how decision trees can overfit the training data and how pruning techniques can improve generalization.

Tasks

1. Train a decision tree classifier without restricting its depth. Record training and testing accuracy.
2. Train multiple models by varying the following hyperparameters:
   - max_depth = 3, 5, 7
   - min_samples_split = 2, 5, 10
3. Compare the performance of these models on both training and testing datasets.
4. Plot the relationship between tree depth and accuracy.
5. Apply cost-complexity pruning using different values of ccp_alpha.
6. Identify the optimal model based on test performance.
7. Explain the concept of overfitting in decision trees and describe how pruning helps reduce it.
8. Discuss which hyperparameter had the greatest impact on model performance.